Supplementary Information for Why Does Life Exist?

Michael Timothy Bennett

Overview

This Supplementary Information provides proofs and experimental details for the main text. The goal is to keep the main paper readable while still giving a complete mathematical spine.

Mathematically, the argument has three parts. Propositions 1 and 2 show that under the novelty model, future function is monotone in buffer size. The theorem then shows that conditioning on survival increases the expectation of any monotone rarity score built from that future function. That second step is generic. The first step is the substantive content of the paper. Proposition 3 extends the analysis to non-exchangeable priors by decomposing any prior into an exchangeable shadow and a residual, and bounding the conditions under which buffer-size monotonicity can reverse. A parametric block-structured family then makes the failure mode explicit, with a closed-form threshold for ordering reversal.

Proofs

Uniform ignorance survival probability

The proposition is restated from the main text.

Proposition 1 (Uniform ignorance survival law)**.** *Let U be a finite unobserved region and let $B \subseteq U$ be a buffer. Let $S \subseteq U$ be drawn uniformly from 2^U. Then*

$$\mathbb{P}(S \subseteq B) = \frac{2^{|B|}}{2^{|U|}} = 2^{|B|-|U|}.$$

Proof. There are $2^{|U|}$ subsets of U in total. There are $2^{|B|}$ subsets of B. The event $S \subseteq B$ holds exactly when S is one of those $2^{|B|}$ subsets. So $\mathbb{P}(S \subseteq B) = \frac{2^{|B|}}{2^{|U|}}$. □

Explanation. If you are ignorant, every possibility is equally likely. You survive exactly when the real outcome lands inside the buffer. The probability is just a counting ratio between the size of the buffer relative to the set of all possibilities.

Exchangeable monotonicity

The proposition is restated from the main text.

Proposition 2 (Exchangeable monotonicity)**.** *Fix a finite set U and let P be an exchangeable distribution over subsets $S \subseteq U$. Let $B \subseteq U$. Then $P(S \subseteq B)$ is a nondecreasing function of $|B|$.*

Proof. Because P is exchangeable, it can be described by the distribution of the random size $K := |S|$. Condition on $K = k$. Given $K = k$, the set S is uniformly distributed over the $\binom{|U|}{k}$ subsets of size k. So

$$P(S \subseteq B \mid K = k) = \begin{cases} \frac{\binom{|B|}{k}}{\binom{|U|}{k}} & \text{if } k \le |B|\,, \\ 0 & \text{if } k > |B|\,. \end{cases}$$

For fixed k, the ratio $\frac{\binom{|B|}{k}}{\binom{|U|}{k}}$ is nondecreasing in $|B|$. Now take expectation over K. A nonnegative weighted sum of nondecreasing functions is nondecreasing. So $P(S \subseteq B)$ is nondecreasing in $|B|$. □

Explanation. Given the number of surprises, survival is a combinatorics ratio that grows when the buffer grows.

Law of increasing functional information

The theorem is restated from the main text.

Theorem 1 (Law of increasing functional information under novelty selection)**.** *Assume an exchangeable novelty prior P. Fix an observed outcome set $\alpha \subseteq L$. Let Π be the set of policies correct on α. Draw π uniformly from Π. Let $S \subseteq U = L \setminus \alpha$ be the novelty set drawn from P. Define $B_\pi = \mathrm{Ext}(\pi) \cap U$. Define degree of future function $F(\pi) = P(S \subseteq B_\pi)$. Define*

$$M(\pi) = \{\pi' \in \Pi \mid F(\pi') \ge F(\pi)\}, \qquad I(\pi) = -\log_2\left(\frac{|M(\pi)|}{|\Pi|}\right).$$

Assume moreover that the survival event has positive probability under the joint law of π and S. Condition on survival, meaning the event $S \subseteq B_\pi$. Then

$$\mathbb{E}\big[I(\pi) \mid S \subseteq B_\pi\big] \;\ge\; \mathbb{E}\big[I(\pi)\big],$$

with strict inequality whenever $F(\pi)$ is not constant over Π.

Proof. Write $\Pi = \{\pi_1, \ldots, \pi_n\}$. Order the policies so that

$$q_j := F(\pi_j) = P(S \subseteq B_{\pi_j})$$

is nondecreasing in j. Now define $I_j := I(\pi_j)$. Because $I(\pi)$ is defined by ranking policies by $F(\pi)$, the sequence (I_j) is also nondecreasing in j.

Let

$$\bar{q} := \frac{1}{n}\sum_{j=1}^{n} q_j, \qquad \bar{I} := \frac{1}{n}\sum_{j=1}^{n} I_j.$$

By assumption, $\bar{q} > 0$. Because the population is uniform on Π,

$$\mathbb{E}[I(\pi)] = \bar{I}, \qquad \mathbb{E}[I(\pi) \mid S \subseteq B_\pi] = \frac{\sum_{j=1}^{n} q_j I_j}{\sum_{j=1}^{n} q_j} = \frac{1}{\bar{q}} \cdot \frac{1}{n}\sum_{j=1}^{n} q_j I_j.$$

So it is enough to show that

$$\frac{1}{n}\sum_{j=1}^{n} q_j I_j \ge \bar{q}\,\bar{I}.$$

Using the standard pairwise covariance identity,

$$\frac{1}{n}\sum_{j=1}^{n} q_j I_j - \bar{q}\,\bar{I} = \frac{1}{2n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}(q_i - q_j)(I_i - I_j).$$

Every term on the right-hand side is nonnegative because both sequences are nondecreasing in the same order. Hence

$$\frac{1}{n}\sum_{j=1}^{n} q_j I_j \geq \bar{q}\,\bar{I},$$

which gives

$$\mathbb{E}[I(\pi) \mid S \subseteq B_\pi] \geq \mathbb{E}[I(\pi)].$$

This is the stated inequality.

If $F(\pi)$ is not constant over Π, then there exist indices i, j with $q_i < q_j$. Because I ranks policies by F, this implies $I_i < I_j$. So at least one term in the double sum is strictly positive, and the inequality is strict. □

Explanation. Survival multiplies each policy's weight by a factor that grows with the survival probability. Functional information is a monotone function of that survival probability inside the viable set. So the mean functional information cannot decrease under survival conditioning, and it rises strictly whenever survival probabilities differ across viable policies.

Robustness to non-exchangeable priors

The results above assume exchangeable novelty. When the prior P over novelty sets is not exchangeable, survival depends on which outcomes sit in the buffer, not just how many. The following proposition quantifies how far from exchangeability P must be before the buffer-size ordering can reverse.

Proposition 3 (Exchangeable decomposition and reversal bound)**.** *Let U be a finite set and let P be any distribution over subsets $S \subseteq U$. Define the exchangeable shadow $\bar{P}$ by*

$$\bar{P}(A) \;=\; \frac{1}{|U|!}\sum_{\sigma \in \mathrm{Sym}(U)} P\big(\sigma^{-1}(A)\big) \qquad \textit{for every event } A \subseteq 2^U,$$

where $\mathrm{Sym}(U)$ is the symmetric group on U and $\sigma^{-1}(A) = \{\sigma^{-1}(T) : T \in A\}$. Then:

(a) $\bar{P}$ is exchangeable, and $\bar{P}(S \subseteq B)$ depends only on $|B|$ for any $B \subseteq U$.

(b) For any buffer $B \subseteq U$,

$$\big|P(S \subseteq B) - \bar{P}(S \subseteq B)\big| \;\leq\; d_{\mathrm{TV}}(P, \bar{P}),$$

where d_{TV} denotes total variation distance.

(c) If $|B_1| > |B_2|$ and the survival ordering reverses under P, meaning $P(S \subseteq B_2) > P(S \subseteq B_1)$, then

$$d_{\mathrm{TV}}(P, \bar{P}) \;>\; \tfrac{1}{2}\big[\bar{P}(S \subseteq B_1) - \bar{P}(S \subseteq B_2)\big].$$

Proof. Part (a). For any permutation $\tau \in \mathrm{Sym}(U)$ and any event $A \subseteq 2^U$,

$$\bar{P}\big(\tau^{-1}(A)\big) = \frac{1}{|U|!} \sum_{\sigma \in \mathrm{Sym}(U)} P\big(\sigma^{-1}(\tau^{-1}(A))\big) = \frac{1}{|U|!} \sum_{\sigma \in \mathrm{Sym}(U)} P\big((\tau\sigma)^{-1}(A)\big) = \bar{P}(A),$$

because $\sigma \mapsto \tau\sigma$ is a bijection on $\mathrm{Sym}(U)$. So $\bar{P}$ is invariant under permutations of U, hence exchangeable. For any two buffers $B, B' \subseteq U$ with $|B| = |B'|$, there exists $\tau \in \mathrm{Sym}(U)$ mapping B onto B'. Then $\{S : S \subseteq B'\} = \tau(\{S : S \subseteq B\})$, so $\bar{P}(S \subseteq B') = \bar{P}(S \subseteq B)$ by invariance.

Part (b). By definition, total variation distance satisfies $|P(E) - Q(E)| \leq d_{\mathrm{TV}}(P, Q)$ for every event E. Applying this with $Q = \bar{P}$ and $E = \{S : S \subseteq B\}$ gives the stated bound.

Part (c). Write $\Delta_i = P(S \subseteq B_i) - \bar{P}(S \subseteq B_i)$ for $i = 1, 2$. By part (b), $|\Delta_i| \leq d_{\mathrm{TV}}(P, \bar{P})$ for each i. The reversal assumption gives

$$P(S \subseteq B_2) > P(S \subseteq B_1),$$

which after substitution becomes

$$\bar{P}(S \subseteq B_2) + \Delta_2 > \bar{P}(S \subseteq B_1) + \Delta_1.$$

Rearranging,

$$\Delta_2 - \Delta_1 > \bar{P}(S \subseteq B_1) - \bar{P}(S \subseteq B_2).$$

Because $|B_1| > |B_2|$, Proposition 2 gives $\bar{P}(S \subseteq B_1) \geq \bar{P}(S \subseteq B_2)$, so the right-hand side is nonnegative. Now

$$\Delta_2 - \Delta_1 \leq |\Delta_2| + |\Delta_1| \leq 2\, d_{\mathrm{TV}}(P, \bar{P}).$$

Combining the two inequalities,

$$2\, d_{\mathrm{TV}}(P, \bar{P}) > \bar{P}(S \subseteq B_1) - \bar{P}(S \subseteq B_2),$$

which gives the stated bound after dividing by two. □

Explanation. Part (a) constructs the exchangeable baseline by symmetrising. Part (b) says the deviation from that baseline is controlled by total variation distance. Part (c) says buffer-size monotonicity can reverse only when the structured prior deviates from its exchangeable shadow by more than half the survival gap between the two buffer sizes under the shadow. If the novelty prior is close to exchangeable, the survival ordering tracks buffer size. If it is far from exchangeable, then survival depends on which specific outcomes the buffer covers, not just how many.

A parametric family of structured priors

The following example makes the failure mode from Proposition 3 concrete. Partition U into m blocks of equal size $b = |U|/m$. Define two component priors over subsets $S \subseteq U$. The exchangeable component draws S uniformly from all $2^{|U|}$ subsets of U. The structured component draws S uniformly from all 2^m unions of whole blocks. The mixture prior with parameter $\rho \in [0, 1]$ is

$$P_\rho \;=\; (1 - \rho)\, P_{\mathrm{exch}} \;+\; \rho\, P_{\mathrm{block}}.$$

Under the exchangeable component, Proposition 1 gives $P_{\mathrm{exch}}(S \subseteq B) = 2^{|B| - |U|}$. Under the structured component, S is a union of whole blocks, so $S \subseteq B$ exactly when every block present

in S is fully contained in B. Let $c(B)$ denote the number of blocks fully covered by B. Then $P_{\text{block}}(S \subseteq B) = 2^{c(B)-m}$, because there are $2^{c(B)}$ unions of covered blocks out of 2^m total.

The mixture survival probability is therefore

$$P_\rho(S \subseteq B) \;=\; (1-\rho)\, 2^{|B|-|U|} \;+\; \rho\, 2^{c(B)-m}.$$

One term depends only on buffer size. The other depends only on block coverage. The whole tension between size and composition is visible in this single expression.

Reversal threshold. Consider two buffers with $|B_1| > |B_2|$ but $c(B_1) < c(B_2)$. The first buffer is larger but covers fewer whole blocks. Under P_ρ, the survival ordering reverses when

$$\rho\left(2^{c(B_2)-m} - 2^{c(B_1)-m}\right) \;>\; (1-\rho)\left(2^{|B_1|-|U|} - 2^{|B_2|-|U|}\right).$$

Writing $\delta_{\text{size}} = 2^{|B_1|-|U|} - 2^{|B_2|-|U|}$ for the exchangeable size advantage and $\delta_{\text{block}} = 2^{c(B_2)-m} - 2^{c(B_1)-m}$ for the structured coverage advantage, the critical mixing weight is

$$\rho^* \;=\; \frac{\delta_{\text{size}}}{\delta_{\text{size}} + \delta_{\text{block}}}.$$

For $\rho < \rho^*$, the larger buffer survives better. For $\rho > \rho^*$, the buffer with more block coverage survives better despite being smaller.

Interpretation. When $\rho = 0$, survival is a function of size alone. Once $\rho > 0$, size alone is no longer sufficient, and ordering reversals can occur whenever a smaller buffer covers more whole blocks. The threshold ρ^* gives a closed form answer to how much structure is needed to override the size advantage. In environments where novelty arrives in correlated clusters (spatial patches, phylogenetic radiations, technological paradigm shifts), the block prior is a minimal model of that clustering. The reversal threshold tells you how strongly clustered novelty must be before alignment with clusters matters more than raw option volume.

Experimental details and additional numbers

Experiment A is generated from a minimal novelty only toy world included with this draft package. The script `run_all_experiments.py` writes all numerical results into `results_macros.tex` and renders Figures 1 and 2. The broader code and theory stack is documented in the technical appendices release [1].

Experiment E is intentionally different. It is a minimal analytic damage model that separates two routes to persistence in a stable environment. One route is to avoid damage by having few essential parts. The other route is to repair damage when you have many essential parts. It does not require enumerating a finite world. The script `run_all_experiments.py` included with this draft generates Figure 2.

Selection rules

I compared two rules.

1. Weakness selection chooses a policy in Π that maximises $|B_\pi|$.

2. Random selection chooses uniformly from Π.

Metric glossary

I use five simple metrics.

- Survival probability is the probability that a policy that is correct on the observed outcomes remains correct after novelty is drawn from the stated prior.
- Log survival in bits is $\log_2$ of that probability. Larger is better.
- The Bayes optimal policy is the policy that maximises survival probability under the stated novelty prior.
- Regret in bits is the log survival gap between the Bayes optimal policy and the policy chosen by the rule. Smaller is better.
- Optimality rate is the fraction of trials where the rule chose a Bayes optimal policy. Larger is better.

Explanation. Regret tells you how many bits of survival probability you leave on the table. Optimality rate tells you how often you hit the best possible policy under the model.

Key summary numbers

Experiment A compares weakness selection and random choice in fully enumerated toy worlds under novelty.

- Toy worlds 200 and total trials 4000.
- Outcome catalogue size $|L| = 48$ and feature length $d = 10$.
- Base commitments 2 and observed outcomes $|\alpha| = 4$.
- Uniform novelty. Weakness improves mean log survival by 1.674 bits relative to random choice.

Experiment E is the damage and repair model from the main text.

- Horizon 40 time steps.
- Per part failure probability $p = 0.020$ per time step.
- Repair budget $r = 1$ failed parts per time step.
- Simple system has $k = 3$ essential parts and no repair.
- Complex system has $k = 12$ essential parts. I evaluate it with no repair and with bounded repair.
- Median survival for the simple system is 12 time steps.
- Median survival for the complex system without repair is 3 time steps.
- Median survival for the complex system with repair is 30 time steps.

References

[1] Michael Timothy Bennett. *Technical Appendices*. Version 5.0. Archived release on Zenodo. Source repository: https://github.com/ViscousLemming/Technical-Appendices. 2025. DOI: 10.5281/zenodo.17164785. URL: https://zenodo.org/records/17164785 (visited on 03/01/2026).